

AZURE CLOUD ENVIRONMENT – VM DEPLOYMENT, STORAGE, MONITORING & TROUBLESHOOTING

Training Project / Practical Assignment

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1. Objective:

The objective of this practical exercise was to create and troubleshoot a basic cloud environment using Microsoft Azure. The exercise involved creating an Azure Virtual Machine, understanding its networking configuration, configuring basic remote access, exploring Azure Storage, examining monitoring metrics, and documenting common troubleshooting scenarios.

2. Azure Environment:

The objective of this practical exercise was to create and troubleshoot a basic cloud environment using Microsoft Azure. The exercise involved creating an Azure Virtual Machine, understanding its networking configuration, configuring basic remote access, exploring Azure Storage, examining monitoring metrics, and documenting common troubleshooting scenarios.

Resource	Name
Resource Group	RG-ITSupport-Practice
Virtual Machine	VM-ITSupport-Practice
Storage Account	itsupportpractice2026
Blob Container	practice-files

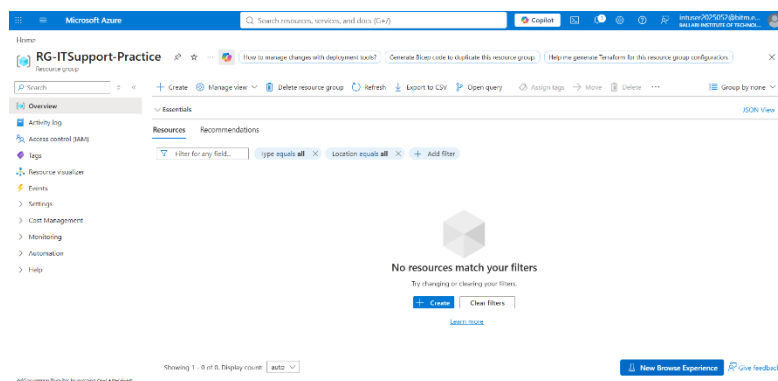


Figure 1: Azure Resource Group containing the practice resources

3. Azure Virtual Machine Creation:

An Azure Virtual Machine (VM) was created as the primary compute resource for the practical environment. A Windows-based VM was deployed through the Azure Portal. The VM was used to practice remote access, examine networking components, and understand basic cloud resource management.

After deployment, the VM Overview page was used to check the VM status and review basic resource information. The VM was also started and stopped during the practical exercise to understand its operational states.

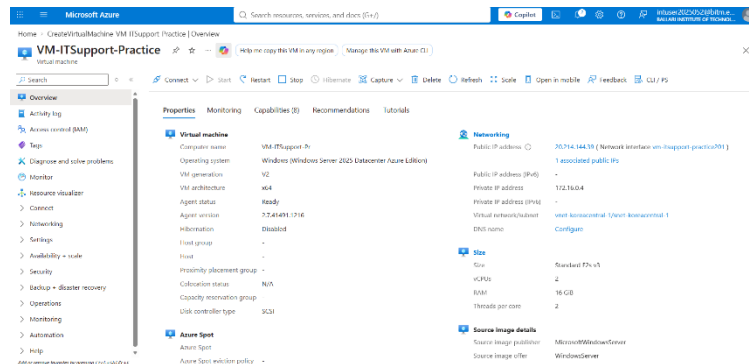


Figure 2: Azure Virtual Machine overview and configuration

4. Networking Configuration:

The networking configuration of the Azure VM was examined to understand how the virtual machine communicates within the Azure environment. The associated Virtual Network (VNet), subnet, Network Interface Card (NIC), private IP address, public IP address, and Network Security Group (NSG) were reviewed.

The Network Security Group associated with the VM was reviewed to examine its network access configuration. The networking configuration was also considered during the troubleshooting process for potential connectivity issues.

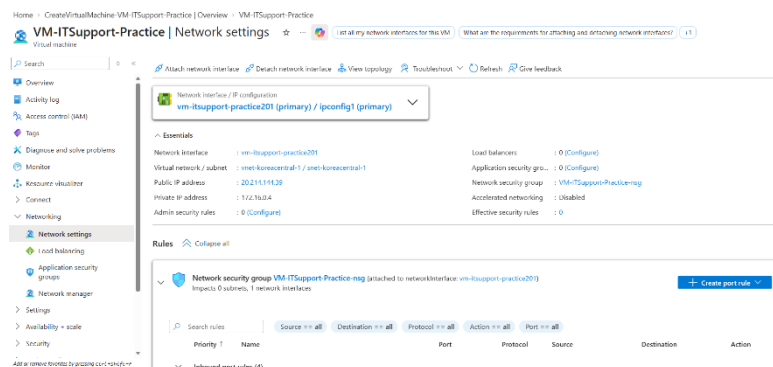


Figure 3: VM networking configuration showing the associated network resources

5. Basic Access Using RDP:

Remote Desktop Protocol (RDP) was used to establish basic remote access to the Windows Azure VM. The VM's public IP address was used with the Windows Remote Desktop Connection client.

After authentication, a successful remote desktop session was established, confirming that the VM could be accessed remotely.

The RDP connection provided practical experience with remotely accessing and managing a cloud-hosted Windows machine.

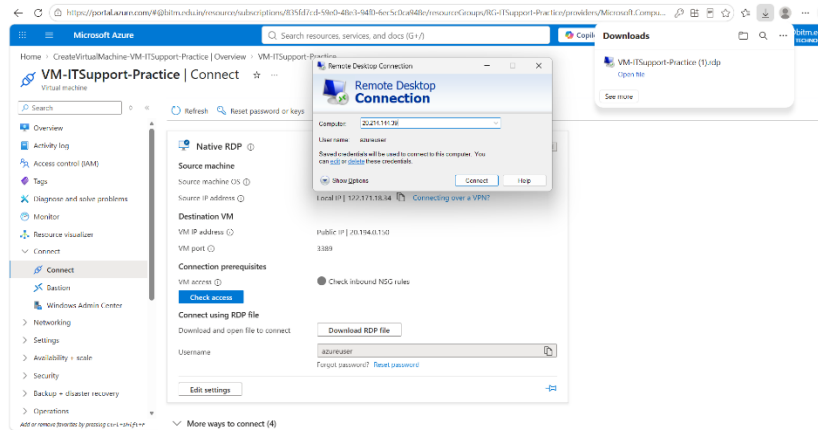


Figure 4: Successful Remote Desktop connection to the Azure VM

6. Azure Storage:

An Azure Storage Account was created to explore cloud-based storage. A private Blob Storage container named practice-files was created within the Storage Account. A sample document, SIMULATED TICKETS.docx, was uploaded to the container and verified after upload.

This exercise demonstrated the basic process of creating cloud storage, organizing files using containers, and uploading files to Azure Blob Storage.

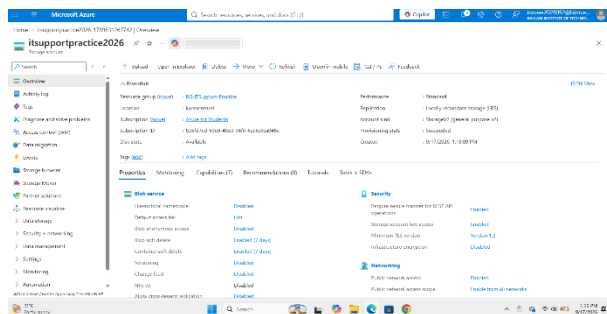


Figure 5: Azure Storage Account and Blob container

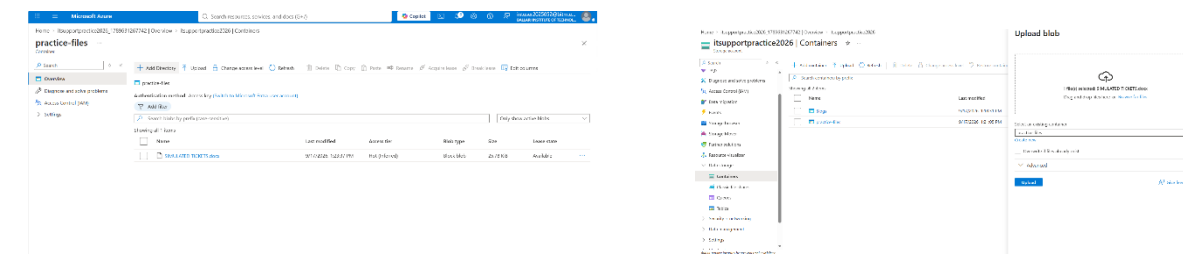


Figure 6: Uploaded file in the practice-files container

7. Monitoring:

Azure Monitor Metrics was used to examine resource activity associated with the Azure environment. The available metrics reviewed during the exercise included Network In Total, Network Out Total, Disk Read Bytes, and Disk Write Bytes. These metrics provided an overview of network and disk activity associated with the virtual machine.

Monitoring metrics can be useful during troubleshooting because unusual network or disk activity may provide additional information about a resource's condition.

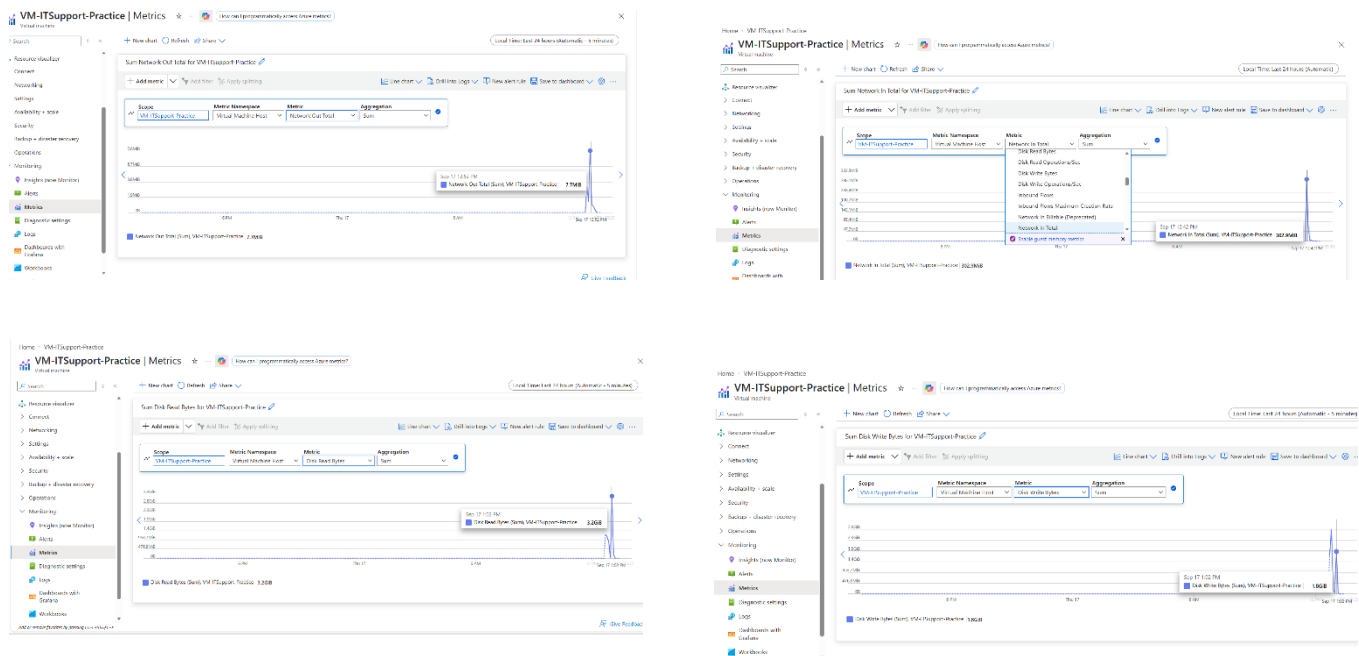


Figure 7: Azure Monitor metrics for examining VM resource activity

8. Troubleshooting Scenarios:

8.1 RDP Connection Failure Due to Outdated Public IP

Problem:

Unable to connect to the Azure Windows Virtual Machine using the previously downloaded RDP file.

Troubleshooting:

The current public IP address displayed on the Azure VM **Connect** page was compared with the IP address stored in the previously downloaded RDP file.

Root Cause:

The RDP file contained the **old public IP address** of the VM. After the VM was stopped/deallocated, its public IP address had changed.

Resolution:

The current public IP address was verified in the Azure Portal, and a new RDP file was downloaded using the updated connection information.

Verification:

The updated RDP configuration was used to establish the remote connection to the VM.

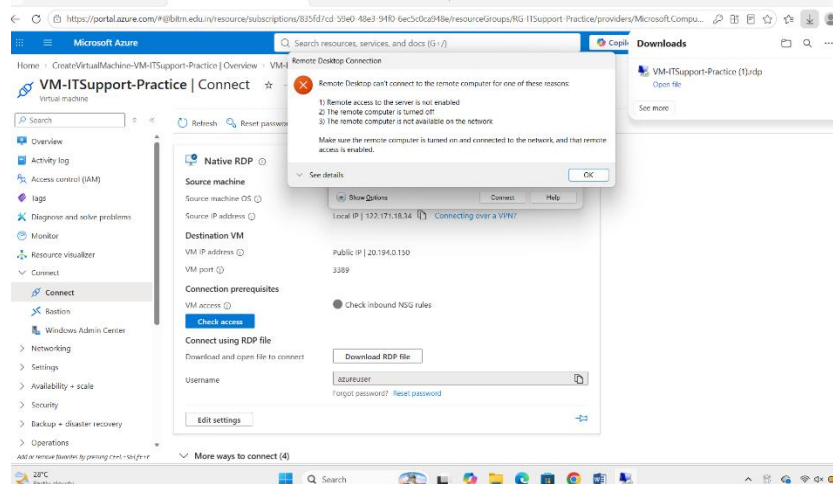


Figure 8: RDP connection error caused by the outdated public IP address.

8.2 Azure VM Network Connectivity Issue

Problem:

The Azure VM is running, but network connectivity or remote access may not work as expected.

Troubleshooting:

The VM's **Virtual Network (VNet)**, **subnet**, **Network Interface Card (NIC)**, **IP configuration**, and **Network Security Group (NSG)** were reviewed. The network access configuration was checked for possible restrictions.

Root Cause:

Incorrect network configuration or an NSG rule restricting the required traffic can cause connectivity problems.

Resolution:

The relevant networking components and access configuration were reviewed to identify and verify the required network connectivity.

Verification:

The VM networking configuration was checked and confirmed to support the required access.

8.3 Azure Blob Storage Upload or Access Issue

Problem:

Unable to upload or access a file in the Azure Blob Storage container.

Troubleshooting:

The **Storage Account**, practice-files container, and storage access configuration were checked. The correct container was verified before attempting the file operation.

Root Cause:

An incorrect container selection or insufficient access permissions can prevent a file from being uploaded or accessed.

Resolution:

The storage configuration and container were verified, and the sample file SIMULATED TICKETS.docx was uploaded successfully.

Verification:

The uploaded file was viewed in the practice-files container to confirm successful storage.